Scaling Keyword Spotting Transformers to Thousand-Class Vocabularies on MSWC

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Abstract. Keyword spotting systems are typically evaluated on 12–35 keywords. We present, to our knowledge, the first systematic evaluation of transformer-based keyword spotting at scales of 1,000 and 6,210 classes on the Multilingual Spoken Words Corpus (MSWC). We implement KWT-2 and benchmark it against five baselines (KNN, SVM, Random Forest, XGBoost, CNN) and the pretrained Audio Spectrogram Transformer (AST). Through systematic error analysis, we identify seven compounding factors and show that architectural corrections improve accuracy from 77.6% to 81.2%. We further fine-tune the Audio Spectrogram Transformer (AST, 87M parameters) and find it achieves only 77.8%—below our from-scratch KWT-2 (2.9M parameters)—revealing a significant AudioSet-to-speech domain mismatch. This study spans 10 model configurations across 3 vocabulary scales.

1. Introduction

Keyword spotting (KWS)—detecting specific spoken words in audio—is a core component of voice assistants, wake-word detection, and voice-controlled devices. Modern voice interfaces rely on KWS as the first stage in a speech processing pipeline: the system must reliably detect a trigger word ("Hey Siri," "Alexa," "OK Google") before activating more expensive automatic speech recognition. Beyond wake-word detection, KWS enables voice-controlled IoT devices, accessibility tools, and hands-free interfaces where the full vocabulary may extend to hundreds or thousands of commands.

The standard benchmark for KWS research is Google Speech Commands V2, which contains 35 English keywords with approximately 85,000 one-second audio clips [1]. While this benchmark has driven significant progress, it represents a constrained evaluation: real-world voice interfaces must handle vocabularies orders of magnitude larger. To our knowledge, no published work has evaluated attention-based KWS models beyond 35 classes, leaving a gap in understanding how these architectures scale.

The Multilingual Spoken Words Corpus (MSWC) [2], published at NeurIPS 2021, spans 340,000 keywords across 50 languages. The English partition has 38,174 words and 5.2 million samples, extracted from Mozilla Common Voice using forced alignment. The original paper benchmarked only 7 keywords: DSCNN at 85.3% and 5-shot transfer at 94.7%. MSWC's scale makes it ideal for studying how KWS models behave when vocabularies grow by two orders of magnitude.

The Keyword Transformer (KWT) [3] achieved 97.7% on Speech Commands using self-attention over temporal audio frames. Unlike CNNs that process local patches, the transformer's self-attention mechanism allows every audio frame to attend to every other frame simultaneously, capturing long-range temporal dependencies. The Audio Spectrogram Transformer (AST) [4], pretrained on AudioSet (2M YouTube clips) and ImageNet, achieved 98.1% on the same benchmark. However, to our knowledge, neither model has been evaluated beyond 35 classes.

We address four questions: (1) How does KWT-2 scale from 31 to 6,210 classes? (2) What factors limit accuracy at thousand-class scale? (3) Can architectural corrections close the gap? (4) Does AudioSet-pretrained AST outperform from-scratch KWT-2 on keyword spotting?

Our contributions:

1. To our knowledge, the first transformer KWS evaluation at 1,000 and 6,210 class scales.
2. Seven compounding errors explaining the 20-point gap between Speech Commands (97.7%) and MSWC (77.6%).
3. Architectural corrections improving accuracy from 77.6% to 81.2%.
4. Evidence that pretrained AST (87M, 77.8%) underperforms from-scratch KWT-2 (2.9M, 81.2%) due to domain mismatch.
5. A scaling study spanning 10 configurations across 3 scales.

2. Related Work

Keyword Spotting Architectures. Deep learning for KWS began with CNNs on MFCC spectrograms [8]. The DSCNN [9] reduced model size through depthwise separable convolutions while maintaining competitive accuracy, becoming the standard edge-deployment baseline.

The Keyword Transformer [3] was the first pure self-attention model for KWS, treating each MFCC time frame as a token in a sequence. KWT-2 uses 12 encoder layers with 2 attention heads and achieves 97.7% on Speech Commands V2 with 5.4M parameters in its largest variant. The key insight is that self-attention allows every audio frame to directly attend to every other frame, capturing temporal dependencies that CNNs can only access through stacked layers.

AST [4] adapts the Vision Transformer to audio by treating

log-mel spectrograms as images, splitting them into 16×16 patches. Pretrained on ImageNet and fine-tuned on AudioSet (2M clips), AST achieved 98.1% on Speech Commands—the highest reported accuracy. However, this performance relies heavily on supervised pretraining rather than the architecture itself.

Pretraining for Audio. Data2Vec pretraining improved KWT by +2–11% in label-deficient settings [6]. SSAST [7] showed 60.9% average improvement from self-supervised pretraining, but noted AST "does not perform well when trained from scratch." The PETL-AST study [13] found linear probing on frozen AST collapsed to 41.78% on Speech Commands, suggesting AudioSet features are poorly aligned with speech. These findings motivated our investigation into whether AudioSet pretraining transfers to large-vocabulary keyword spotting.

MSWC. The original paper [2] benchmarked DSCNN at 85.3% and 5-shot prototypical-network transfer at 94.7% on 7 English keywords. We are not aware of subsequent work evaluating large-vocabulary supervised classification on MSWC, which makes ours, to our knowledge, the first study to test transformers at this scale.

3. Methodology

3.1 Dataset

We use three partitions of MSWC English, summarized in Table 1.

Table 1: Dataset scales (original recordings, before augmentation).

Partition	Classes	Train	Test
Microset (A)	31	17,923	4,553
Mid-scale (B)	1,000	299,700	50,000
Full-scale (C)	6,210	309,900	154,154

The microset uses official speaker-disjoint splits. Experiments B and C cap the number of recordings per keyword. The v1 runs capped each keyword at 500 recordings (1,000 classes) or 200 (6,210 classes) with no augmented copies. The v2 runs capped each keyword at 300 or 50 recordings and added two augmented copies of each (speed perturbation, time shift), tripling the training set. This is why the training sizes in Tables 3 and 4 differ from Table 1. The full English dataset is 33 GB extracted from a 124 GB compressed archive.

3.2 Baselines

We compare KWT-2 against five baselines spanning instance-based, margin-based, tree-based, and neural approaches. KNN and SVM operate on the same flattened MFCC vectors as the tree-based methods and serve as simple distance- and margin-based references.

KNN. k-nearest neighbors with $k = 5$ and Euclidean distance, operating on flattened 13×100 MFCC vectors.

SVM. Support vector classifier with an RBF kernel ($C = 1$, γ scale), same flattened MFCC input.

CNN. Three convolutional blocks (32, 64, 128 filters, 3×3 kernels), batch normalization, ReLU, max pooling, and a fully connected head. 13-bin MFCC input, approximately 150K parameters.

Random Forest. 200 estimators with Gini impurity, operating on flattened 13×100 MFCC vectors (1,300 features per sample).

XGBoost [12]. 200 gradient-boosted trees, max depth 6, learning rate 0.1, `multi:softmax` objective, same flattened MFCC input.

3.3 KWT-2 Architecture

We implement KWT-2 [3]: 12 transformer encoder layers, 2 attention heads, $d_{\text{model}} = 128$, PostNorm (layer normalization after attention and feedforward), GELU activation. The input audio is converted to a spectrogram (MFCC or log-mel depending on version), where each time frame becomes one token. A learnable [CLS] token is prepended, and learnable positional embeddings encode temporal order. The [CLS] token's output representation is passed through the classification head.

v1 Configuration (incorrect): 13-bin MFCC input (100 frames × 13 features), $d_{ff} = 256$, linear classification head ($128 \to n_{\text{classes}}$), SpecAugment [11] only. This was our initial implementation before identifying discrepancies with the published KWT paper.

v2 Configuration (corrected): 40-bin log-mel spectrogram input (100 frames × 40 features), $d_{ff} = 512$ (matching the paper), expansion classification head ($128 \to 512 \to \text{GELU} \to n$) to address the softmax bottleneck [5], SpecAugment with frequency masks of size 7, speed perturbation (0.85×–1.15×), time shift (±100 ms), and Mixup ($\alpha = 0.3$). The expansion head increases the effective rank of the logit distribution, enabling finer discrimination between acoustically similar keywords.

3.4 Audio Spectrogram Transformer (AST)

We fine-tune the publicly released AudioSet+ImageNet-pretrained AST checkpoint[1] (87M parameters) via HuggingFace Transformers. AST treats the log-mel spectrogram as an image, splitting it into 16×16 patches, flattening them into a sequence, and applying a Vision Transformer with 12 layers, hidden size 768, and 12 attention heads. A [CLS] token aggregates the sequence for classification.

AST v1: Freeze 8/12 layers, lr=1e-5 with linear warmup and cosine decay, AudioSet normalization defaults (mean=-4.27, std=4.57), batch size 16, FP16 mixed precision, 30 epochs.

AST v2: Freeze 4/12 layers, lr=1e-4 with 10% warmup and cosine decay, MSWC-specific normalization (mean=-9.21, std=6.85), SpecAugment, same batch size and precision, 30

[1] HuggingFace checkpoint: `MIT/ast-finetuned-audioset-10-10-0.4593`.

epochs.

Critical finding: MSWC audio has a mean log-mel value of -9.21, substantially different from AudioSet's default -4.27. Since the feature extractor normalizes as $(x - \text{mean})/(2 \times \text{std})$, incorrect statistics shift every input by a constant bias. We computed MSWC statistics by sampling 2,000 training files and extracting 128-bin log-mel features at 16 kHz with 10 ms hop length.

AST v3 (failed): We also attempted full fine-tuning (0 frozen layers) with lr=2.5e-4, but this destroyed the pretrained weights entirely, collapsing to 0.1% accuracy. This confirms that aggressive learning rates are destructive for pretrained audio transformers.

3.5 Training Details

All KWT-2 models use AdamW [10] with lr=0.001, weight decay 0.1, cosine annealing with warmup, label smoothing 0.1, and gradient clipping at 1.0. The CNN uses Adam with lr=0.001. KWT-2 experiments ran on an NVIDIA RTX 3060 (12 GB, FP32). AST fine-tuning ran on a Google Colab A100 (40 GB, FP16 mixed precision) with batch size 16.

4. Results

4.1 Microset Results (31 Classes)

Table 2: 31-class microset results.

Model	Accuracy	F1	Time
KNN (k=5)	37.0%	—	<1 s
Random Forest	48.3%	0.47	12 s
SVM (RBF)	61.8%	—	45 s
XGBoost	63.2%	0.62	38 s
CNN (3-block)	86.7%	0.87	6 min
KWT-2 v1	**86.8%**	**0.87**	85 min

Tree-based methods (48–63%) cannot compete with neural approaches (86.7%+), as shown in Table 2. The 23-point gap between XGBoost and the transformer demonstrates that learned hierarchical features are essential for audio classification. Random Forest and XGBoost operate on flattened 1,300-dimensional MFCC vectors with no temporal structure, treating each feature independently. By contrast, the CNN learns local spectral patterns through convolution, and the transformer captures global temporal dependencies through self-attention. The transformer matches the CNN on this small dataset; its advantage emerges with more data at larger scales.

Tree-based methods complete training in under a minute, while the transformer requires 85 minutes on a Mac M1 CPU. For deployment scenarios where accuracy requirements are modest and training speed matters, XGBoost provides a reasonable baseline with minimal computational overhead. However, the 23-point accuracy gap makes neural approaches the clear choice for production keyword spotting.

4.2 v1 Scaling Results

Table 3: KWT-2 v1 scaling (13-bin MFCC).

Scale	Classes	Train	Accuracy
Microset	31	17,923	86.8%
Mid-scale	1,000	499,500	77.6%
Full-scale	6,210	1,085,898	77.6%

As reported in Table 3, accuracy drops 9.2 points from 31 to 1,000 classes and plateaus. With $d_{\text{model}} = 128$, the logit matrix has rank ≤ 128—a softmax bottleneck that caps accuracy once the number of classes exceeds 128.

4.3 Seven Compounding Errors

We identify seven factors that we believe contribute to the 20-point gap (97.7% on Speech Commands versus 77.6% on MSWC). The percentages are rough estimates rather than isolated measurements, and they overlap rather than add up:

1. **Wrong features** ($\sim$4%): 13-bin MFCC versus 40-bin log-mel.
2. **Halved** d_{ff} ($\sim$2–3%): 256 versus the paper's 512.
3. **Softmax bottleneck** ($\sim$5–8%): $d_{\text{model}} = 128$ with 6,210 classes.
4. **No pretraining** ($\sim$5–10%): published 90%+ KWS systems on large-vocabulary speech rely on pretraining [6,7].
5. **Missing augmentation** ($\sim$2%): no speed perturbation or noise.
6. **MSWC data quality** ($\sim$5–7%): force-alignment errors in auto-extracted audio.
7. **Insufficient epochs** ($\sim$1–3%): loss still decreasing at epoch 30.

Items 1–3, 5, and 7 are architectural (fixable). Items 4 and 6 require pretraining or data cleaning.

4.4 Detailed Error Breakdown

To understand the relative importance of each factor, we analyze them in two tiers.

Tier 1: Architectural errors (fixable). Wrong input features account for the largest share. The 13-bin MFCC applies a discrete cosine transform (DCT) that compresses spectral information into 13 coefficients per frame, discarding fine-grained frequency structure. The 40-bin log-mel spectrogram retains all 40 mel-scaled frequency channels, giving the transformer three times more spectral detail per token. For distinguishing acoustically similar words like "nine" and "mine," which differ primarily in nasal formant frequencies around 300 Hz, this additional resolution is critical.

The halved feedforward dimension ($d_{ff} = 256$ instead of 512) reduces the model's capacity to learn nonlinear transformations within each encoder layer. Since the feedforward sublayer is the primary mechanism for learning complex feature interactions (attention captures dependencies, feedforward transforms representations), halving it effectively halves the model's expressiveness.

The softmax bottleneck is perhaps the most theoretically interesting error. With $d_{\text{model}} = 128$, the logit vector for any input must lie in a 128-dimensional subspace. When the number of classes exceeds this dimensionality ($6{,}210 \gg 128$), the model cannot represent a distribution where each class has an independent logit direction. This forces semantically distinct keywords to share logit directions, creating systematic confusion.

Tier 2: External factors (require pretraining or data work). No model in the literature achieves 90%+ on 500+ audio classes without large-scale pretraining. Data2Vec, WavLM, and HuBERT have all demonstrated that speech-specific pretraining is essential for high accuracy at scale. MSWC's data quality—auto-extracted from Common Voice using forced alignment—introduces a noise floor. The original paper reports that short words like "did" have 32% word error rates even among the nearest-neighbor samples.

4.5 v2 Corrections

Table 4: KWT-2 v2 results (corrected architecture).

	Classes	Train	v1	v2
Mid-scale	1,000	899,100	77.6%	**81.2%**
Full-scale	6,210	929,700	77.6%	77.0%

As shown in Table 4, the v2 corrections yield +3.6% on 1,000 classes from architecture alone. The 6,210-class experiment did not improve due to data starvation (50 original recordings per class).

4.6 AST Fine-Tuning Results

Table 5: AST results on 1,000-class MSWC (A100 GPU).

Configuration	Frozen	LR	Accuracy
AST v1 (AudioSet norm)	8/12	1e-5	63.4%
AST v2 (MSWC norm)	**4/12**	**1e-4**	**77.8%**

Table 5 summarizes the two AST configurations we evaluated. Moving from AST v1 to v2 (corrected normalization statistics, four more trainable layers, a higher learning rate, and SpecAugment) yielded +14.4 percentage points—the largest gain in our study. AST v2 learns fast (43.4% at epoch 1 versus KWT-2's 1.3%) but plateaus at 77.8% by epoch 23, overfitting while training loss continues decreasing.

4.7 Complete Results

Table 6 consolidates all 10 model configurations across the three vocabulary scales, alongside the two MSWC paper baselines for context.

Table 6: Complete comparison: 10 configurations, 3 scales.

Model	Scale	Accuracy	Parameters
KNN (k=5)	31	37.0%	N/A
Random Forest	31	48.3%	N/A
SVM (RBF)	31	61.8%	N/A
XGBoost	31	63.2%	N/A
CNN (3-block)	31	86.7%	150K
KWT-2 v1	31	86.8%	1.6M
KWT-2 v1	1,000	77.6%	1.7M
KWT-2 v1	6,210	77.6%	2.4M
KWT-2 v2	**1,000**	**81.2%**	**2.9M**
KWT-2 v2	6,210	77.0%	5.6M
AST v1	1,000	63.4%	87M
AST v2	1,000	77.8%	87M
MSWC DSCNN [2]	7	85.3%	—
MSWC 5-shot [2]	7	94.7%	—

5. Discussion

5.1 The Pretraining Paradox

Our most striking finding: from-scratch KWT-2 (2.9M parameters, 81.2%) outperforms pretrained AST (87M parameters, 77.8%) on 1,000-class MSWC. This 30× parameter and pretraining advantage is not enough to overcome AudioSet's domain gap with isolated keyword speech, qualifying the broader assumption that pretrained audio models transfer well to downstream audio tasks.

AudioSet contains 10-second YouTube clips of general audio: music, nature, machinery. These features are fundamentally different from 1-second speech keywords. AST's pretrained layers encode patterns for dog barks and car horns, not phonetic distinctions between "nine" and "mine." This aligns with the SSAST authors' observation [7] and PETL-AST's finding [13] that frozen AudioSet features collapse to 41.78% on Speech Commands.

This result has practical implications: practitioners building keyword spotting systems should not default to the largest pretrained model available. A task-specific architecture trained from scratch on in-domain data can outperform a general-purpose pretrained model with far more parameters, while also being cheaper to deploy on edge devices where KWS systems typically run.

5.2 Architecture versus Pretraining

Our ablation across v1 and v2 demonstrates a clear hierarchy. Input feature representation provides the largest single improvement: switching from 13-bin MFCC to 40-bin log-mel spectrograms recovers approximately 4% by preserving spectral fine structure that the DCT compression in MFCC discards. The log-mel spectrogram retains all 40 frequency channels, allowing the transformer to learn frequency-specific attention patterns.

The expansion classification head and correct feedforward dimensions contribute approximately 3% combined. The softmax bottleneck—operating a 128-dimensional model

with 6,210 output classes—is a well-characterized limitation [5]. Our expansion head ($128 \to 512 \to \text{GELU} \to n$) increases the effective rank of the logit distribution.

Data augmentation (SpecAugment, speed perturbation, Mixup $\alpha = 0.3$) adds approximately 2%. SpecAugment prevents reliance on specific frequency bands; Mixup regularizes decision boundaries. Total: +3.6% from architecture alone, while AudioSet pretraining provides no net benefit.

5.3 The Normalization Discovery

The +14.4% gain from AST v1 to v2 shows how sensitive pretrained transformers are to their fine-tuning setup. Normalization is a likely contributor: the feature extractor normalizes as $(x - \mu)/(2\sigma)$, so when the mean is wrong by 5 units (-9.21 versus -4.27), every input is shifted by a constant offset. However, v2 also unfroze four more layers, raised the learning rate, and added SpecAugment, so this experiment does not isolate the effect of normalization alone. A controlled ablation that changes only the normalization statistics would be needed. Regardless, any fine-tuning of AST on a new audio domain should begin by computing dataset-specific normalization statistics.

5.4 Scaling Behavior and the Softmax Bottleneck

The 77.6% plateau at both 1,000 and 6,210 classes demonstrates a representational bottleneck. With $d_{\text{model}} = 128$, the logit matrix $W \in \mathbb{R}^{128 \times n}$ has rank ≤ 128—far fewer than the 6,210 classes it must distinguish. The expansion head partially addresses this. For further scaling, increasing d_{model} to 256 or adopting metric learning (e.g., ArcFace loss) may be necessary.

5.5 AST Training Dynamics

AST v2 shows rapid initial learning: 43.4% at epoch 1 (versus KWT-2's 1.3%). However, accuracy plateaus at 77.8% by epoch 23 while training loss continues decreasing (5.40 to 1.18), indicating overfitting. The frozen early layers produce features optimized for general audio, limiting the model's ability to learn speech-specific patterns.

5.6 Comparison with Published Baselines

Our KWT-2 v2 at 81.2% on 1,000 classes is not directly comparable to the MSWC paper's DSCNN at 85.3% on 7 classes, but the result suggests that transformers can maintain reasonable accuracy at 143× the vocabulary scale.

5.7 Limitations

Our XGBoost accuracy (63.2%) is reported from a single training run rather than averaged across multiple seeds; we expect ±1–2% variance under repeated runs. We explored only two AST configurations, and they differ in several settings at once, so the contribution of each change is not isolated; exhaustive hyperparameter search might improve results. The error estimates in Section 4.3 are approximate. We did not implement MSWC's data-quality filtering (k-means anomaly detection), which improved DSCNN by 2.7%. MSWC's auto-extracted audio imposes an accuracy floor regardless of model choice.

6. Future Work

Speech-Specific Pretraining. The primary lesson from our AST experiments is that the pretraining domain matters more than scale. Speech-specific pretraining using Data2Vec [6] on LibriSpeech (960 hours) or WavLM would provide representations aligned with phonetic structure. Bovbjerg and Tan [6] showed +2–11% gains; we expect similar or larger improvements on MSWC.

MSWC Data Quality Filtering. The original paper showed that k-means anomaly detection improved DSCNN from 85.3% to 88.0%. Given that data quality accounts for an estimated 5–7% of our gap, this represents a significant opportunity.

Cross-Lingual Transfer. MSWC covers 50 languages, many with severely limited labeled data. Training KWT-2 on English (the highest-resource language) and fine-tuning on low-resource languages such as Sakha, Dhivehi, or Maltese with 5–50 samples per keyword would test whether transformer keyword spotting generalizes across languages. This directly competes with the MSWC paper's 94.7% 5-shot baseline, which used a prototypical network approach. If successful, cross-lingual transfer could enable voice interfaces for underserved languages without collecting thousands of labeled examples per keyword.

Scaling d_{model}. Our softmax bottleneck analysis suggests that increasing d_{model} from 128 to 256 or 512 would directly address the rank limitation at 6,210 classes. This would approximately double or quadruple training time and memory, but could be offset by reducing encoder layers from 12 to 6 or 8, maintaining total parameter count while improving the output representation capacity. An alternative approach is to adopt metric learning losses such as ArcFace that operate on normalized embeddings rather than raw logits, circumventing the bottleneck entirely.

7. Conclusion

We present, to our knowledge, the first evaluation of transformer-based keyword spotting at 1,000 and 6,210 class scales on MSWC. Our systematic analysis reveals that naïve application of KWT-2 with 13-bin MFCC features plateaus at 77.6% on 1,000 classes—a 9.2-point drop from the 31-class microset and a 20-point drop from published Speech Commands benchmarks. Through careful error analysis, we identified seven compounding factors responsible for this gap and showed that correcting five architectural errors—switching to 40-bin log-mel features, restoring the paper's feedforward dimension, adding an expansion classification head to mitigate the softmax bottleneck, incorporating augmentation with SpecAugment and Mixup, and extending training—improves accuracy from 77.6% to 81.2%, a +3.6% absolute gain.

Our most significant finding is the pretraining paradox. A from-scratch KWT-2 with 2.9 million parameters (81.2%) outperforms a pretrained Audio Spectrogram Transformer with 87 million parameters (77.8%) by 3.4 percentage points. This demonstrates that AudioSet pretraining—despite representing orders of magnitude more compute and data—does not transfer effectively to isolated keyword speech. The domain mismatch between general audio (music, environmental sounds, machinery) and 1-second spoken words is too large for the pretrained representations to overcome. This finding is consistent with observations by the SSAST [7] and PETL-AST [13] authors and has direct practical implications: practitioners building KWS systems should prioritize correct architecture and in-domain training over transfer from mismatched pretraining corpora.

We also found that the fine-tuning setup of pretrained audio transformers is critical and easy to get wrong. Moving from AST v1 to v2, which corrected the normalization statistics (mean=-9.21 instead of AudioSet's -4.27) while also unfreezing four more layers and raising the learning rate, yielded a +14.4% accuracy gain—the single largest improvement in our experimental campaign. Isolating the share due to normalization is left to future work, but computing dataset-specific statistics should be the first step of any fine-tuning experiment.

Our complete scaling study spans 10 model configurations (KNN, SVM, Random Forest, XGBoost, CNN, KWT-2 v1 at three scales, KWT-2 v2 at two scales, and AST at two configurations) across 3 vocabulary scales (31, 1,000, and 6,210 classes), with 7 identified error factors providing the most extensive scaling analysis of attention-based keyword spotting that we are aware of.

All code, results, and documentation are available at `https://github.com/HamdanDarei/mswc-keyword-spotting`.

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